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SMART FOREST MONITORING SYSTEM

Software Requirement Engineering

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Project submitted

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1. PROBLEM DOMAIN

1.1 Background to the Problem

Forests play a crucial role in maintaining ecological balance, supporting biodiversity, and sustaining life on earth. However, forests worldwide are increasingly facing serious threats due to illegal activities such as poaching, logging, robbery, criminal activities, and smuggling. Additionally, natural disasters like wildfires and landslides further endanger forest ecosystems and the communities that rely on them. The lack of an efficient monitoring system has led to a rise in these problems, making it challenging for forest departments and government agencies to respond promptly and effectively.

The root cause of these problems is the inadequate surveillance and real-time monitoring of forest areas. Traditional monitoring methods rely heavily on manual patrolling, which is often inefficient and insufficient for covering large forest areas. The absence of advanced technological solutions makes it difficult to detect and prevent illegal activities and natural disasters before they cause significant damage. This problem is critical to address as it poses a severe threat to wildlife conservation, environmental sustainability, and public safety.

1.2 Solution to the Problem

The proposed solution for this project is a Forest Monitoring System that makes forest monitoring easier, faster, and more efficient by using modern technology. The system will use modern technologies like IoT sensors, GPS tracking, AI-based cameras, drones, and mobile applications to collect real-time information from forest areas. This solution is suitable because traditional forest monitoring methods are slow, manual, and difficult to manage in large forest regions. With the help of technology, the system can quickly detect illegal activities, forest fires, or other environmental threats and send alerts to the responsible authorities instantly.

The main purpose of this software is to improve forest protection, safety, and management. It will help forest departments monitor forest conditions more efficiently and reduce problems such as illegal logging, poaching, and wildfire damage. The system will also support tourists and local communities by providing safety alerts and useful information through a mobile application. The software aims to save time, reduce human effort, and improve decision-making using real-time data and automated monitoring features.

This solution is feasible because many of the required technologies are already available and widely used around the world. IoT devices, GPS systems, drones, and AI tools can be integrated together to create an effective monitoring platform. The system also supports the business objective by

improving efficiency, reducing monitoring costs in the long term, and helping authorities take quick action during emergencies.

Several existing studies and software solutions are already available in this field. Some systems use GIS and satellite imaging for forest monitoring, while others use drones and IoT devices for data collection. However, many of these solutions do not provide real-time monitoring, automated alerts, or AI-based threat prediction. Traditional patrolling systems are also still used in many areas, but they require a lot of manpower and are less effective for large forests. The proposed Forest Monitoring System improves these existing solutions by combining multiple technologies into one smart and integrated platform for better forest management.

2. SOLUTION DESCRIPTION

2.1 System features

Functional Requirements:

1. User Registration and Authentication

1.1 The system will enable secure login using usernames and passwords. Users will be categorized as administrators, forest officers, or researchers.

- During registration, users must provide email, contact number and select their role.
- Passwords will be encrypted and stored using secure hashing algorithms.
- The login module will verify credentials against stored values in the database.

1.2 If login fails multiple times, the system will trigger a security mechanism that sends a One Time Password (OTP) to the user's registered email.

- Users must verify this OTP before resetting the password.
- The OTP will be time-bound (valid for 2 minutes) and will expire thereafter.

1.3 Multi-Factor Authentication will be enforced by requiring an additional OTP sent to the registered mobile/email after the initial password verification step.

- This ensures identity verification beyond a password.
- Users cannot access the system until the second step is completed.

1.4 Role-Based Access Control will dynamically assign permissions.

- Admins can manage sensors, users, and all system configurations.
- Forest officers can view environmental data, manage alerts, and contact emergency services.
- Researchers can access reports, historical data, and analytics modules.

Priority Level: High

Requirement: Robust, secure, and role-based user access with modern security practices.

2. Dashboard

2.1 The dashboard will present live environmental data like temperature, humidity, soil moisture, and rainfall collected from IoT sensors.

- Data will be fetched via Message Queuing Telemetry Transport(MQTT)/HyperText Transfer Protocol (HTTP) protocols and pushed to the frontend using WebSockets or server-sent events for real-time updates.

- Information will be displayed in digital tiles with live indicators (example: blinking green for active, red for critical values).

2.2 Wildlife movement detected by motion sensors or camera traps will generate instant pop-up notifications with location and time info.
- These will be shown as real-time toast messages, along with map markers indicating the movement zone.

2.3 Smoke or fire sensors will trigger dashboard alerts.
- A flashing red icon will appear, sound alarms may activate, and an alert banner will show detailed info like coordinates, sensor ID, and timestamp.

2.4 Historical trends will be represented using **interactive graphs and charts** (example: line charts, heatmaps).
- Users can filter data by date range, sensor type, or zone.

Priority Level: High

Impact: Helps users make rapid decisions based on real-time and historical insights.

3. Sensor Management

3.1 Admin users can perform (Create, Read, Update, Delete) operations on sensors through a dedicated interface.
- A sensor registration form will allow input of type, location, threshold parameters, and description.
- Each sensor will be assigned a unique ID.

3.2 The system will ping sensors periodically (example: every 5 minutes) to check their status.
- Online sensors will show a green icon; offline ones will appear grayed out with a last seen timestamp.

3.3 If a sensor fails to report data or goes offline, the system will raise an automatic alert.
- These alerts will be logged and shown on the dashboard and via email.

Priority Level: High

Impact: Maintains data integrity and ensures system reliability.

4. Alerts and Notifications

4.1 The system will use rule-based triggers to detect and alert about events such as:
- Forest fires (via smoke or temperature spikes)

- Illegal activities (motion detection in restricted hours or zones)
- Wildlife movement (triggered by animal-tagged sensors or camera traps)
- Extreme weather (sudden pressure drops or rainfall spikes)

1.3 Alerts will pass through a verification module that cheaks for false positives using sensor correlation or admin review (optional for low-priority alerts).

2.3 Verified alerts will be delivered using multiple channels:

- Email and SMS (via 3rd-party services like Twilio)
- Dashboard notification center
- Push notifications on the mobile app

4.4 Critical alerts (e.g., fire or poaching detection) will be forwarded to emergency services or forest police with location and severity details.

Priority Level: High

Impact: Improves coordination and emergency response across multiple teams.

5. Data Logging and Reporting

5.1 All data from sensors will be logged in a centralized, cloud-based database (e.g., Firebase, PostgreSQL) with timestamped entries.

5.2 Users can generate custom reports:

- Filterable by sensor type, date, location, or event type
- Exportable in CSV and PDF formats for ease of sharing

5.3 AI-based predictive models will run daily on historical data.

- These models will detect trends and predict events like potential wildfires, flooding, or sensor failure zones.
- Results will be visualized and shared via dashboard widgets.

Priority Level: High

Impact: Supports proactive decisions based on predictive insights.

6. Tourist Safety Features

6.1 The mobile app will include real-time GPS tracking using Google Maps API or OpenStreetMap.

- Tourists can follow designated safe trails with live positioning.

6.2 The system will notify tourists of hazards using geofencing:

- When a user enters a restricted or dangerous zone, the app vibrates and displays a warning popup.

6.3 The app will feature an **SOS button** for emergencies.

- When triggered, it will send location and ID to forest authorities and the nearest emergency center.

Priority Level: High

Impact: Enhances safety and ensures timely rescue if necessary.

7. Advanced Monitoring Technologies

7.1 Drones will be deployed at set intervals to patrol the forest autonomously.

- They will stream live video to the dashboard and record it for future review.
- Drones can be manually controlled or follow pre-programmed routes.

7.2 Thermal imaging systems on drones or ground units will operate during nighttime.

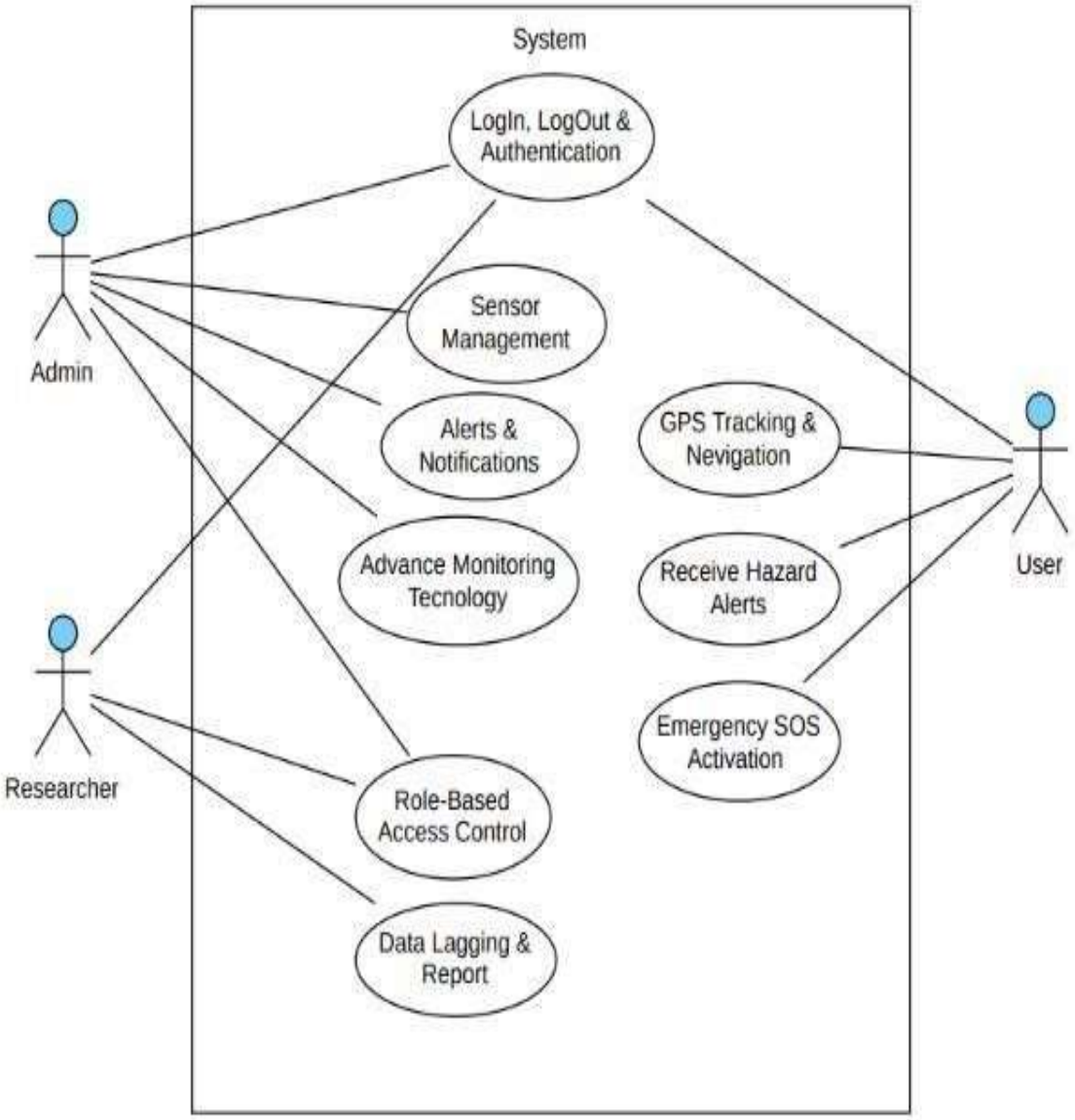
- They'll identify unusual heat signatures (from fires or unauthorized human presence).
- Images will be processed and alerts generated automatically when patterns are matched.

Priority Level: High

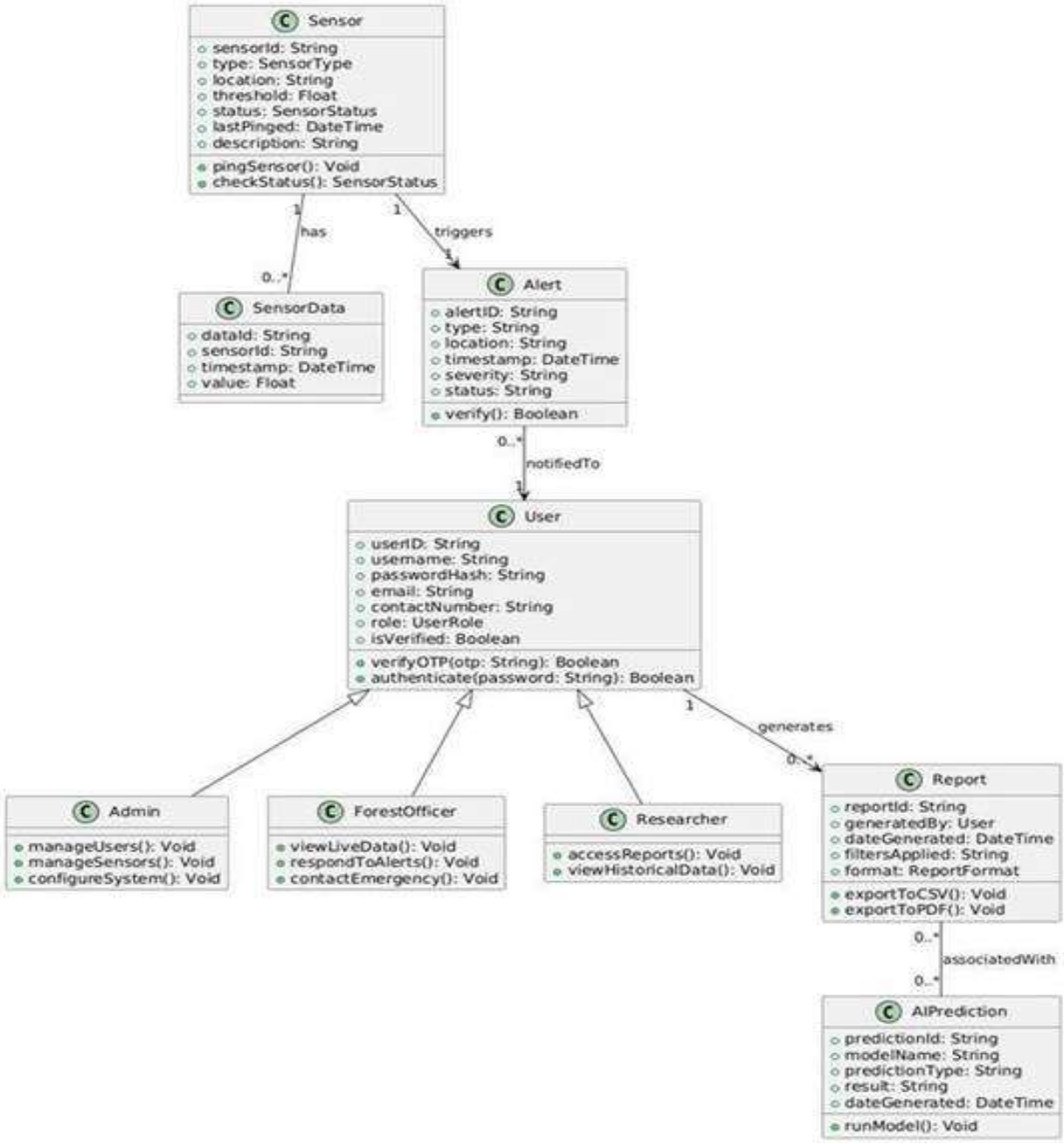
Impact: Adds advanced surveillance capabilities for maximum forest protection.

2.2 UML Diagrams

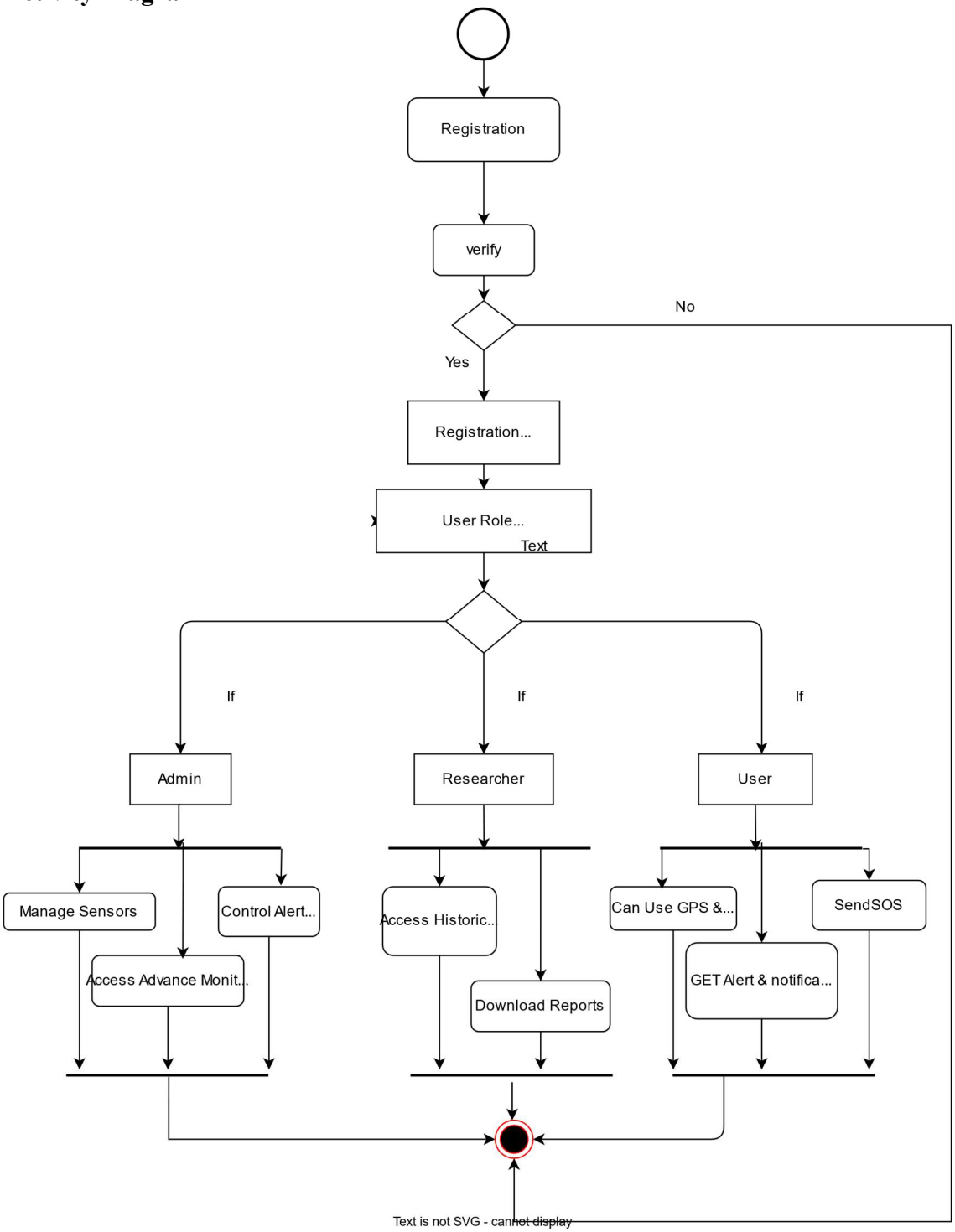
1. Use case Diagram:



2. Class Diagram



3. Activity Diagram



3. Social Impact

The proposed Smart Forest Monitoring System will create a positive impact on society in several ways. The system will help protect forests, wildlife, and natural resources by detecting illegal activities such as poaching, illegal logging, and forest fires at an early stage. As a result, environmental damage can be reduced and biodiversity can be preserved more effectively.

This project will also improve public safety. Tourists and local people visiting forest areas will receive real-time alerts about dangerous situations like wild animal movement, fire risks, or restricted zones. The emergency SOS feature can help authorities respond quickly during accidents or emergencies.

In addition, the system will support government agencies and forest departments by providing real-time data and automated monitoring. This will reduce manual work, save time, and help officials make faster and better decisions for forest management and conservation.

The project can also increase public awareness about environmental protection and encourage people to participate in forest conservation activities. Overall, the Smart Forest Monitoring System will contribute to environmental sustainability, public safety, and better management of forest resources for future generations.

4. Development Plan with Project Schedule

The Smart Forest Monitoring System project will follow the Agile Software Development Life Cycle (SDLC) model. The project will be divided into multiple sprints where each sprint will focus on specific features, testing, feedback collection, and improvements. The Agile approach will help the development team work more efficiently, manage changing requirements, and deliver the project in a more organized way.

The project schedule below is designed based on realistic software development practices where planning, design, development, testing, and deployment require proper time management.

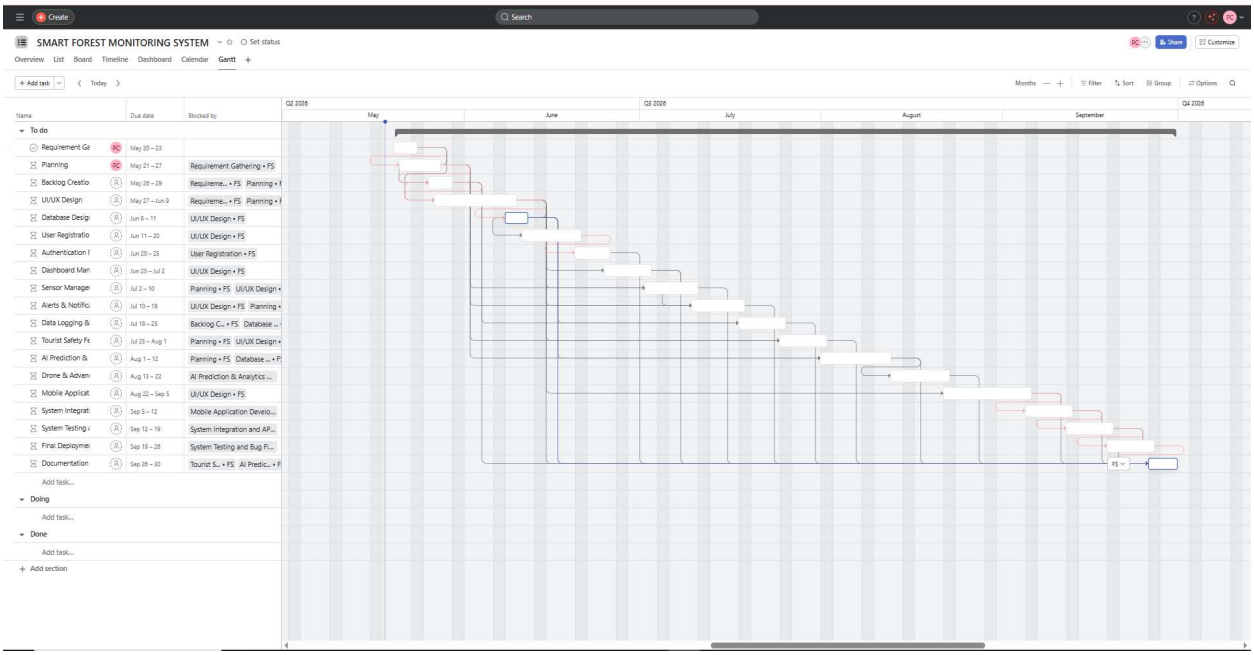


Fig- Gantt chart

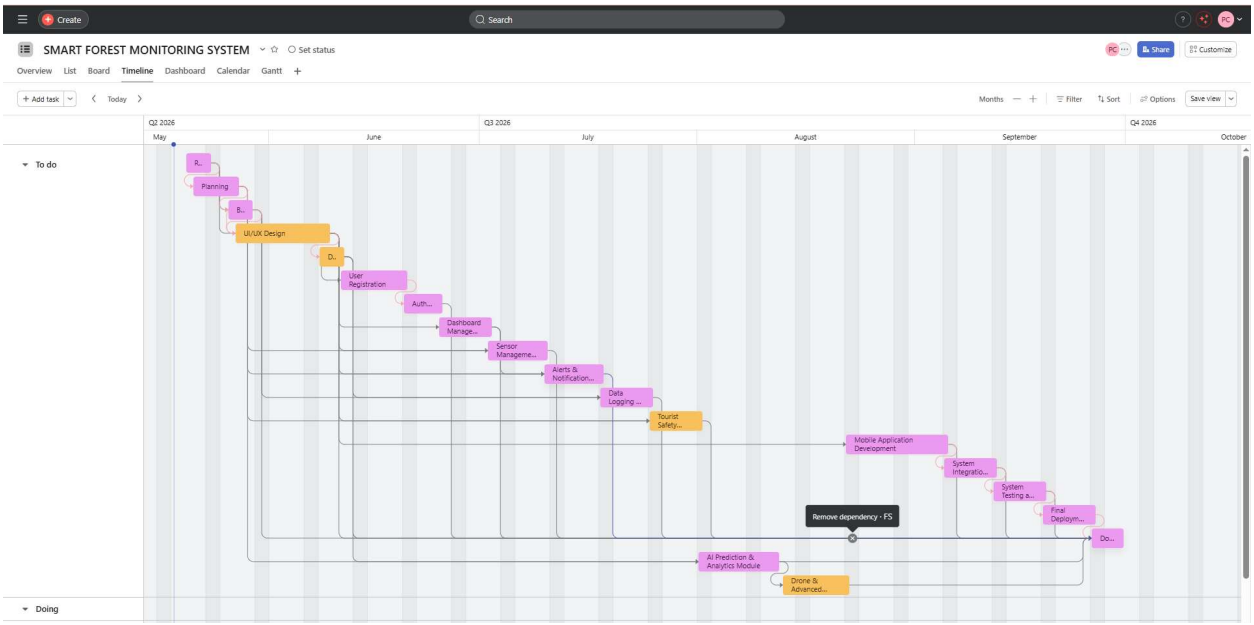


Fig- Project Timeline

5. Marketing Plan

The marketing plan for the Smart Forest Monitoring System will focus on increasing awareness about smart forest protection technologies and encouraging government organizations, environmental agencies, and conservation groups to use the system. Since the project is related to environmental safety and forest management, the marketing strategy will mainly focus on building trust, demonstrating the effectiveness of the system, and creating long-term partnerships. The marketing activities will be divided into short-term, long-term, and continuous plans to ensure steady growth and business opportunities.

• Short-Term Plan

In the initial stage, the project will mainly focus on introducing the Smart Forest Monitoring System to potential users and organizations. Awareness campaigns will be conducted through social media platforms, technology seminars, workshops, and project presentations. The team will create a professional website containing project information, system features, benefits, and demo videos to help people understand how the system works.

Live demonstrations and prototype presentations will also be arranged for forest departments, environmental organizations, universities, and research groups. These demonstrations will help show the practical benefits of the system such as real-time monitoring, fire detection, wildlife tracking, and emergency alerts. Social media marketing on platforms like Facebook, YouTube, and LinkedIn will also be used to reach a larger audience and create public awareness about forest conservation and smart monitoring technologies.

• Long-Term Plan

In the long term, the project will aim to build partnerships with government agencies, NGOs, wildlife conservation organizations, and private environmental companies. The system can gradually be introduced in different forest areas, wildlife sanctuaries, and national parks. Long-term contracts and service agreements can also be developed for system maintenance and monitoring support.

As technology continues to improve, additional features such as advanced AI-based prediction systems, drone automation, thermal imaging, and mobile application improvements can be added to make the system more effective and competitive. The project may also expand internationally in the future, especially in countries where forest protection and disaster prevention are major concerns.

The long-term goal is to establish the Smart Forest Monitoring System as a reliable, modern, and widely used environmental monitoring solution that can support both forest safety and wildlife conservation.

- ## Continuous Plan

To keep the project active and successful, continuous updates and improvements will be provided based on user feedback and technological advancements. Regular software updates, security improvements, and maintenance services will help maintain system performance and reliability.
The project team will continue organizing awareness programs, environmental campaigns, workshops, and training sessions to educate people about the importance of forest conservation and smart monitoring systems. Customer support services will also be maintained to help users solve technical problems and improve user experience.
In addition, feedback from forest officers, researchers, tourists, and environmental organizations will be collected regularly to improve system features and introduce new functionalities based on real-world needs. Continuous marketing through social media, websites, and public events will help maintain public interest and attract new clients over time.
Overall, this marketing plan will help create awareness, build trust among organizations, and generate business opportunities while contributing to environmental protection, public safety, and sustainable forest management.

6. Cost and Profit Analysis

6.1 Development Cost:

The development cost includes software development, system design, hardware integration, testing, cloud services, and maintenance expenses required to build the Smart Forest Monitoring System.

Development Components	Estimated Cost (BDT)
Software Development	150,000
UI/UX Design	20,000
Database & Cloud Services	15,000
IoT Sensors & Hardware	60,000
Drone Integration	50,000
AI & Analytics Development	35,000
Testing & Maintenance	20,000
Mobile Application Development	40,000
Miscellaneous Cost	10,000
Total Development Cost	**4,00,000 BDT**

6.2 Marketing Cost :

Marketing costs are required to promote the project and attract clients, government organizations, and investors.

Marketing Activities	Estimated Cost (BDT)
Social Media Marketing	15,000
Website Development & Hosting	10,000
Promotional Campaigns	20,000
Workshops & Seminars	15,000
Client Meetings & Presentations	10,000
Total Marketing Cost	**70,000**

6.3 Total Estimated Cost :

Category	Amount (BDT)
Total Development Cost	4,00,000
Total Marketing Cost	70,000
Overall Estimated Cost	**4,70,000**

6.4 Total Estimated Cost :

The Smart Forest Monitoring System has strong business potential because forest departments, environmental organizations, and government agencies require modern monitoring solutions for forest protection and disaster prevention.

- Software installation charges
- Monthly subscription services
- System maintenance fees
- Cloud monitoring services
- Premium AI analytics features
- Government and private sector partnerships

Estimated yearly income from clients and services can reach around 7,00,000 BDT to 10,00,000 BDT depending on project deployment and client acquisition.

6.5 Investment Opportunity :

The Smart Forest Monitoring System has strong investment potential because the need for smart forest protection and environmental monitoring systems is increasing rapidly. Government agencies, forest departments, and environmental organizations can use this system to improve forest safety, reduce illegal activities, and respond quickly during emergencies.

The project uses modern technologies such as AI, IoT sensors, GPS tracking, cloud monitoring, and drone surveillance, which make the system more effective and valuable than traditional monitoring methods. The project can generate profit through software installation, maintenance services, subscriptions, and monitoring solutions.

Since environmental protection is becoming more important worldwide, this project has good future business opportunities and can become a profitable long-term investment for investors.

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